Qn	Key	guide
1	**D**	$mg - U = ma$ $mg - ma = U$ $m(g - a) = U$ $m = \frac{U}{g - a} = \frac{2.40}{9.81 - 4.50} = 0.451977$ kg To determine uncertainties, $\frac{\Delta m}{m} = \frac{\Delta U}{U} + \frac{\Delta(g - a)}{(g - a)} = \frac{0.06}{2.4} + \frac{0.05}{9.81 - 4.50} = 0.034416$ $\Delta m = 0.032564 \times 0.363068 = 0.015555$ kg $= 0.02$ kg (1sf) Hence, final answer is (0.45 ± 0.02) kg
2	**C**	The general equation is $F = kx \rightarrow x = \frac{F}{k}$